

5. Refrigeration

(R-1)

Vapor compression refrigeration cycle:

Example

Consider a 300 kJ/min refrigeration system that operates on an ideal vapor-compression refrigeration cycle with R 134a as working fluid. The refrigerant enters the compressor as saturated vapor at 140 kPa and is compressed to 800 kPa. Show the cycle on a T-s diagram with respect to saturation lines, and determine:

- the quality of refrigerant at the end of the throttling process,
- the coefficient of performance, and
- the power input into the compressor.

Solution

VCR cycle

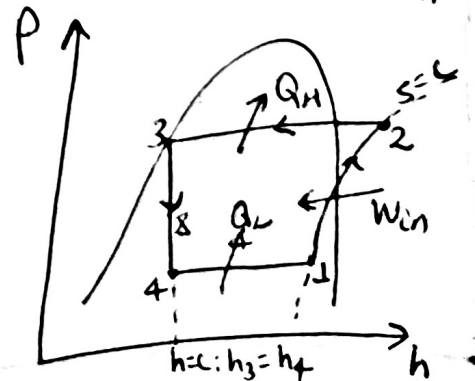
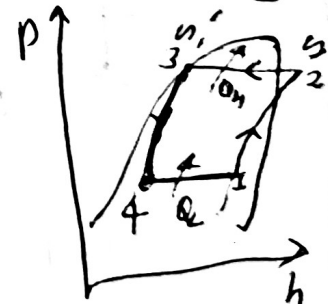
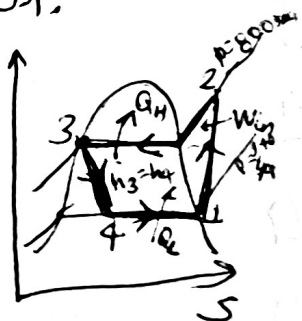
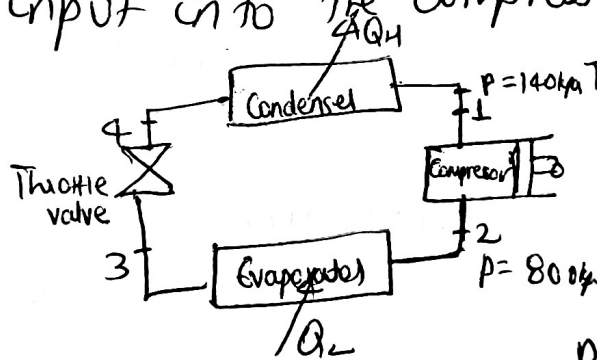
Refrigerant: R 134a

$$\begin{aligned} \text{Power} = \dot{W}_{\text{net}} &= 300 \text{ kJ/min} \\ &= 300 \text{ kJ/min} \times \frac{1 \text{ min}}{60 \text{ s}} \\ &= \underline{\underline{5 \text{ kJ/s}}} \end{aligned}$$

Compressor inlet. Sat. vapor at 140 kPa
Compressor exit : 800 kPa

Required

- Quality at end of throttling process.
- COP_R
- $\dot{W}_{\text{in}}$

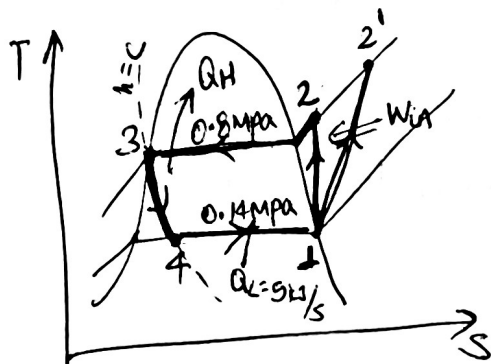


(R-2)

Assumptions

- Steady operating condition exists.
- kinetic & potential energy changes are negligible.

Analysis



States: Using refrigerant table for R-134a
Table A-11, A-12, A-13

State 1: h_1 using $p = 0.14 \text{ MPa}$ Table A-12

$$h_1 = 236.04 \text{ kJ/kg}, s_1 = 0.9322 \text{ kJ/kg}\cdot\text{K}$$

State 2:

$$s_2 = s_1 = 0.9322 \text{ kJ/kg}\cdot\text{K}$$

Superheated table for refrigerant

Table A-13

$$h_2 = 272.05 \text{ kJ/kg}$$

State 3:

saturated liquid, $p = 0.8 \text{ MPa}$

$$h_3 = h_f = 93.42 \text{ kJ/kg}$$

State 4:

Throttling process

$$h_3 = h_4$$

$$h_4 = 93.42 \text{ kJ/kg}$$

Determine x_4 (Quality): at $p = 0.14 \text{ MPa}$

$$h_4 = h_f + x_4 h_{fg}$$

$$h_f = 27.08 \text{ kJ/kg}$$

$$h_{fg} = 212.08 \text{ kJ/kg}$$

$$x_4 = \frac{h_4 - h_f}{h_{fg}} = \frac{93.42 - 27.08}{212.08}$$

$$x_4 = 0.3217$$

$$\Rightarrow a = x_4 = 0.3217$$

$$b. \text{COP}_R = \frac{Q_L}{W_{\text{net}}}$$

$$= \frac{h_1 - h_4}{h_2 - h_1} = \frac{236.04 \text{ kJ/kg} - 93.42 \text{ kJ/kg}}{272.05 \text{ kJ/kg} - 236.04 \text{ kJ/kg}}$$

$$= 3.96$$

$$c. W_{\text{net}} = \dot{m}(h_2 - h_1) \Rightarrow \text{Find } \dot{m} \text{ from } \dot{Q}_L = \dot{m}(h_1 - h_4) \Rightarrow \dot{m} = 0.0351 \text{ kg/s}$$

$$W_{\text{net}} = 1.26 \text{ kW}$$